Event-Time Distribution Modeling for Robust EEG-Based Reaction-Time Prediction

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Keywords: robust EEG decoding, event-time distribution modeling, behavioral decoding, temporal segmentation

Abstract

Reaction time (RT) is an event latency, not only a scalar label attached to an EEG window. We study trial-wise RT prediction from stimulus-locked Healthy Brain Network EEG under cross-subject and cross-release shift, and reformulate the task as event-time distribution modeling: the model predicts a posterior over response latencies and the scalar RT estimate is read out by expectation.

We compare compact scalar regressors, temporal-readout controls, and standardized external EEG architectures trained from scratch under the same preprocessing. Event-time CE and CE+time supervision yield lower R11 nRMSE point estimates than scalar regression and temporal-readout controls, while time-only and Wasserstein-only ablations show that the gain is not explained by a soft-argmax head alone. EventNLL is evaluated as a principled latent event-time likelihood in which observed RT is a noisy realization of an unobserved event time; Gaussian-mixture and hazard/survival variants are compact probabilistic extensions rather than separate main contributions.

Scalar nRMSE is only part of the evidence. Posterior geometry reveals differences in width, target-aligned mass, mode-mean disagreement, and coverage that are hidden by scalar readout error. A shifted-crop diagnostic shows partial event-localizer behavior while also revealing that fixed-window training does not solve shift equivariance. Under a release-separated protocol using R1–R8 for model fitting, R9–R10 for development-stage calibration and diagnostics, and R11 for final reporting, the calibrated CE+time event-time segmentation model reaches 0.935573 nRMSE on R11 [0.918136, 0.954309], compared with 0.945938 [0.930153, 0.963795] for the strongest direct-regression baseline. The original hidden challenge evaluation is reported only as secondary benchmark context.

1 Introduction

Electroencephalography (EEG) is widely used for measuring human brain activity because it is noninvasive, comparatively inexpensive, and provides millisecond-scale temporal resolution. This temporal resolution makes EEG especially attractive for decoding behavioral variables that unfold over time, including reaction times, response preparation, and event-related dynamics. At the same time, EEG decoding remains difficult in heterogeneous datasets because task-relevant signals are low-SNR and vary substantially across subjects, sessions, and recording conditions.

Motivated by the success of foundation models in language and vision, the EEG community has recently seen rapid growth in large-scale EEG pretraining and generalist backbones that aim to learn transferable, task-agnostic representations from heterogeneous EEG corpora. Examples include transformer-based pretraining such as EEGPT (Wang et al., 2024), large multi-dataset tokenization and masking strategies such as LaBraM (Jiang et al., 2024) and EEGFormer (Chen et al., 2024), state-space model backbones such as EEGMamba (Wang et al., 2025), and large-scale cross-setup pretraining explicitly targeting montage and recording variability such as REVE (Ouahidi et al., 2025). Autoregressive pretraining has also been explored for generalist EEG modeling (Yue et al., 2025). Overall, these efforts suggest that large-scale self-supervised pretraining can reduce reliance on task-specific labels.

Despite this progress, robust EEG decoding remains challenging in practice due to low signal-to-noise ratio (SNR) and strong distribution shifts. EEG recordings reflect mixtures of neural sources through volume conduction and are easily contaminated by non-neural artifacts such as eye movements, facial and neck muscle activities, cardiac signals, and environmental noise. EEG is also highly nonstationary across time and context: changes in alertness, movement, task engagement, and electrode impedance can substantially shift the data distribution even within a single session. Crucially for machine learning, there is substantial inter-subject variability driven by anatomy, development, cognitive strategy, and recording conditions. Together, these factors often yield models that perform well within-subject or within-session but degrade sharply when evaluated across subjects, sites, or dataset releases.

In this work, we argue that a practical route to robustness under EEG heterogeneity is to make the temporal structure of the prediction target explicit. We focus on trial-wise reaction-time prediction from stimulus-locked EEG. Reaction time is not merely a scalar label attached to an entire window; it is the latency of a behavioral event following stimulus onset. Treating this latency as ordinary scalar regression discards temporal structure that is naturally present in the task. We therefore reformulate reaction-time prediction as event-time distribution modeling. Given the low-SNR and cross-subject/cross-release shift, we deliberately prioritize lightweight models and controlled objective-design comparisons rather than introducing a larger backbone.

We evaluate this formulation on the reaction-time prediction task introduced by the EEG Foundation Challenge (Aristimunha et al., 2025; EEG Challenge, 2025), using curated releases of the Healthy Brain Network EEG dataset (Shirazi et al., 2024). The task is based on the contrast change detection (CCD) paradigm and requires predicting trial-wise response time from 2 s stimulus-locked EEG windows; our main analysis uses a labeled release-separated protocol over HBN releases.

For reaction-time decoding, we replace direct scalar regression with event-time distribution modeling. Models predict a probability distribution over possible response times within the post-stimulus window and are converted back to scalar RT predictions through an expectation readout. We study soft Gaussian-label segmentation losses, scalar readout ablations, Wasserstein/CDF matching, and a latent event-time likelihood in which the observed RT is a noisy observation of an unobserved response-relevant event time. This formulation supports label-consistent temporal jitter and yields uncertainty-sensitive distributional summaries for posterior-geometry analysis. The goal is formulation-first rather than architecture-first: we ask whether aligning the objective with the event-time nature of RT yields lower held-out RT prediction error even when the underlying EEG models remain lightweight.

Our experiments compare the event-time formulation against compact direct-regression models and standardized external EEG architectures under the same release-separated protocol. The obtained results support the idea that for low-SNR EEG reaction-time prediction, objective design and temporal inductive bias can be at least as important as moderate increases in model capacity within the tested regime.

Our contributions are:

1. We reformulate EEG-based reaction-time prediction as event-time distribution modeling rather than ordinary scalar regression.

2. We evaluate compact direct-regression baselines, temporal-readout regression, and standardized external EEG architectures under a release-separated HBN protocol with R11 subject-bootstrap confidence intervals.

3. We show through loss ablations and five-seed robustness checks that distributional event-time supervision matters beyond a soft-argmax readout alone.

4. We formulate EventNLL as a principled latent event-time likelihood for RT and evaluate compact Gaussian-mixture and hazard/survival extensions as probabilistic alternatives.

5. We analyze posterior geometry and shifted-crop behavior to distinguish scalar accuracy from event-time localization and uncertainty behavior.

2 Related Work

2.1 Robust EEG Decoding under Distribution Shift

A central obstacle for EEG decoding is that the data distribution changes substantially across subjects, sessions, and recording setups. Long before the recent wave of large-scale EEG pretraining and generalist backbones, the BCI community developed methods aimed at reducing this shift through more stable feature spaces and explicit alignment. A prominent example is Riemannian EEG decoding, which represents trials by spatial covariance matrices and uses the geometry of symmetric positive definite matrices to stabilize cross-subject and cross-session classification (Barachant et al., 2012). Related transfer learning work has emphasized distribution alignment and data reuse across subjects or sessions to reduce calibration requirements (Jayaram et al., 2016; Zanini et al., 2018; He and Wu, 2020). Together, these methods highlight a recurring principle: robustness often benefits when learning is constrained by inductive biases that suppress nuisance variability.

Deep learning approaches to robust EEG decoding have imported similar ideas from domain adaptation, including adversarially learned domain-invariant representations (Ganin et al., 2016) and moment-matching objectives such as correlation alignment (Sun and Saenko, 2016). In parallel, self-supervised and large-scale EEG pretraining has emerged as another route to transferability, with contrastive or masked objectives used to learn reusable representations across tasks, subjects, and hardware (Kostas et al., 2021). This direction has recently expanded into multi-dataset EEG foundation models and generalist backbones (Wang et al., 2024; Jiang et al., 2024; Chen et al., 2024; Wang et al., 2025; Ouahidi et al., 2025; Yue et al., 2025).

Our work is complementary to both alignment-centric transfer learning and large-scale pretraining. Rather than proposing a new invariant backbone or an explicit domain-matching loss, we focus on a task-driven source of structure: the temporal nature of the behavioral target. Reaction time is not merely a scalar label attached to an EEG window; it is the latency of a behavioral event following stimulus onset. We therefore reformulate RT prediction as event-time distribution modeling, in which the model predicts a distribution over possible response latencies and the scalar RT estimate is obtained by an expectation readout. The reaction-time task introduced by the EEG Foundation Challenge provides a standardized large-scale setting for studying this question under cross-subject and cross-release shift (EEG Challenge, 2025; Aristimunha et al., 2025). We use it as a test bed for objective design and temporal inductive bias, not as a study of pretraining itself: foundation-style architectures in our experiments are trained from scratch and serve as controlled architecture baselines.

2.2 Reaction-Time Decoding as Event-Time Prediction

Reaction time is inherently temporal: even when evaluation uses a single scalar per trial, the supervised quantity is an event time within a post-stimulus window. This view is closely related to single-trial EEG and ERP work showing that trial-to-trial latency variability can carry behaviorally meaningful structure. ERP-image analyses, for example, sort single-trial epochs by reaction time and reveal stimulus-locked and response-locked components that can be blurred by conventional averaging (Jung et al., 1998). Reviews of single-trial ERP latency estimation further emphasize that intra-subject latency variability, especially in the P3 range, is linked to behavioral variability and can substantially affect averaged ERP morphology (Ouyang et al., 2017). This motivates treating the RT label not only as a trial-level scalar, but as the latency of a response-relevant event whose timing varies across trials.

Supervised RT prediction from single-trial EEG has also been studied directly, but is usually formulated as scalar regression or coarse discretized classification from hand-crafted or learned EEG features (Chowdhury et al., 2020). Our formulation keeps the same supervised prediction setting while changing the output space: instead of directly regressing RT, the model predicts a posterior distribution over candidate response times.

This event-time view is also compatible with computational models of decision time. Sequential-sampling models, especially the diffusion decision model, explain RT distributions through latent evidence accumulation, decision boundaries, and non-decision time components (Ratcliff and McKoon, 2008). Human EEG studies have identified related accumulation-to-bound signals, including centro-parietal dynamics whose buildup rate and boundary crossing relate to perceptual decision time (O'Connell et al., 2012; Kelly and O'Connell, 2013). We do not fit a cognitive process model or infer drift-rate parameters; instead, we adopt the weaker premise that observed RT is a noisy behavioral realization of a latent response-relevant event time.

2.3 Distributional Supervision, Latent Event Times, and Calibration

From a machine-learning perspective, our approach can be viewed as a distributional version of scalar RT regression. Label distribution learning replaces point labels with soft distributions over labels and is particularly natural when neighboring labels share metric structure, as neighboring time bins do in a post-stimulus RT window (Geng, 2016). Time-to-event modeling provides a related language: classical survival analysis treats event time as the central target (Cox, 1972), and modern deep models can directly parameterize discrete event-time distributions (Lee et al., 2018). Although our benchmark does not involve clinical survival outcomes or censoring, the discrete event-time view is useful because it makes timing explicit and supports probabilistic supervision over response latency.

Because our model outputs a posterior over response time, calibration and posterior geometry are part of the modeling problem rather than only implementation details. Recent neural time-to-event work emphasizes calibration and uncertainty for event-time distributions (Chapfuwa et al., 2023), and evaluation frameworks for individual survival distributions provide useful terminology for assessing whether predicted distributional mass, coverage, and uncertainty are meaningful (Haider et al., 2020). In our setting, scalar RT is obtained by the posterior expectation, but the full event-time posterior remains available for diagnosis: its width, entropy, target-aligned mass, mode-mean disagreement, and coverage can distinguish objectives that appear similar under scalar nRMSE.

Our event-time formulation therefore bridges three lines of work: single-trial EEG analyses of response-locked latency variability (Jung et al., 1998; Ouyang et al., 2017), computational accounts of reaction-time distributions (Ratcliff and McKoon, 2008; O'Connell et al., 2012; Kelly and O'Connell, 2013), and modern distributional supervision for labels and event times (Geng, 2016; Lee et al., 2018; Chapfuwa et al., 2023; Haider et al., 2020). The resulting model remains lightweight, but its objective is aligned with the temporal structure of the behavioral target.

3 Dataset and Evaluation

3.1 Reaction-Time Task and EEG Recordings

We use the reaction-time prediction task introduced by the NeurIPS EEG Foundation Challenge (EEG Challenge, 2025; Aristimunha et al., 2025), based on curated releases of the HBN-EEG dataset (Shirazi et al., 2024). The challenge setting provides a standardized task definition, while the main evaluation in this manuscript uses labeled HBN releases rather than the original leaderboard score. The releases comprise recordings from more than 3,000 participants aged 5–21 collected across a standardized battery of six EEG tasks spanning both passive and active paradigms (Shirazi et al., 2024; EEG Challenge, 2025).

HBN-EEG is distributed in a Brain Imaging Data Structure (BIDS)-compatible format with detailed event annotations using Hierarchical Event Descriptors (HED), enabling reproducible preprocessing and time-locked analyses (Shirazi et al., 2024). In the original acquisition, EEG was recorded in a sound-shielded environment using a high-density 128-channel EGI Geodesic HydroCel system at 500 Hz, referenced at Cz, with an online 0.1–100 Hz bandpass and electrode impedances maintained below 40 kΩ. For the benchmark derivative, recordings are band-pass filtered (0.5–50 Hz) and downsampled to 100 Hz to standardize modeling across releases (EEG Challenge, 2025).

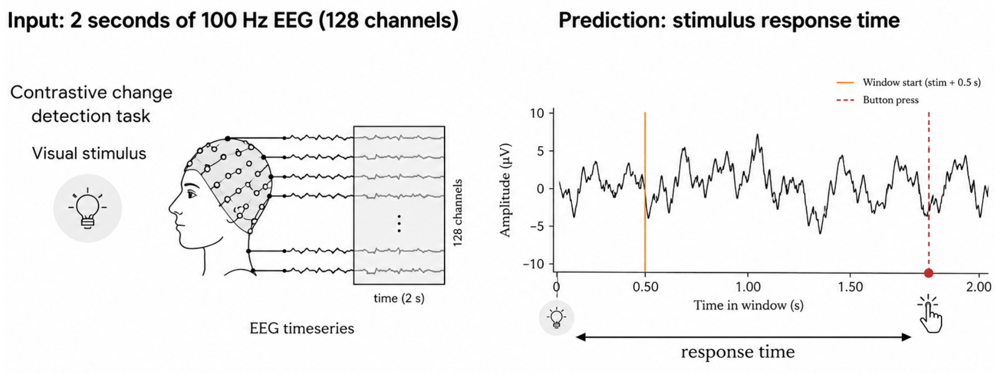

Figure 1: Reaction-time prediction as event-time modeling. Each trial provides a 2 s stimulus-locked EEG window from the CCD paradigm (0.50–2.50 s after stimulus onset; 128 EEG channels at 100 Hz). The observed RT is represented as a smoothed distribution over post-stimulus time bins. The model predicts an event-time distribution, which is converted to a scalar RT prediction using a temperature-scaled soft-argmax readout.

The released derivatives include 129 channels (128 EEG channels plus a reference); throughout this work we train models on the 128 EEG channels and exclude the reference channel from the learnable input.

The CCD task is the active HBN-EEG paradigm used in this study. In each trial, participants viewed two concentric, flickering grating stimuli, one left-tilted and one right-tilted. After a variable delay, the contrast of one stimulus increased, making it perceptually dominant. Participants were instructed to respond as quickly as possible to indicate whether the dominant grating was left- or right-tilted, after which they received immediate accuracy feedback.

Each model input is a fixed-length 2 s EEG segment, represented as 128 EEG channels sampled at 100 Hz and time-locked to the stimulus event. The fixed inference window covers 0.50–2.50 s after stimulus onset (Figure 1). The released target is a scalar RT for each trial, and scalar performance is evaluated using normalized RMSE (nRMSE), defined as RMSE divided by the standard deviation of the true targets (EEG Challenge, 2025). Because the label reflects when a response occurs after stimulus onset, this task naturally motivates modeling RT as an event-time quantity rather than as a static scalar regression target.

3.2 Held-Out Release Evaluation

We evaluate models using a fixed split over HBN releases. Releases R1–R8 are used for fitting the base EEG models. Releases R9–R10 form the development partition used for checkpointing, calibration, and protocol diagnostics. Release R11 serves as the labeled test release for final reporting. Model development, calibration, and selection are fixed before R11 scoring, and main performance results are reported on R11 with subject-level bootstrap confidence intervals. Distribution-aware stacking and the original hidden-release CodaBench evaluation are reported in Appendix A as secondary benchmark context.

For posterior-geometry analyses, we distinguish scalar RT evaluation from event-time support diagnostics. Scalar nRMSE is computed on all R11 trials. Posterior-shape summaries are computed only for trials whose observed RT is representable on the modeled event-time grid. With 200 bins at 100 Hz, the bin centers span 0.50–2.49 s after stimulus onset; the plotted and inference window covers the corresponding 0.50–2.50 s interval. This affects 15,164 of 15,751 R11 trials; 587 trials fall below the modeled support and are excluded only from posterior-geometry summaries, not from scalar performance reporting. Appendix Figure 9 visualizes this support diagnostic.

3.3 Input Normalization and Augmentation

All neural models receive the same input normalization. For each 2 s trial window and each EEG channel, we subtract that channel's temporal mean within the window and divide by its temporal standard deviation within the same window. This transformation is computed independently within each input window and is applied identically across models.

Unless otherwise stated, training uses a fixed mild augmentation recipe. Channel dropout is applied with probability 0.25 and removes at most 30% of channels. Temporal cutout is applied with probability 0.25 and masks a contiguous 10–50 sample interval, corresponding to at most 0.5 s in a 2 s, 100 Hz window. Gaussian noise is applied with probability 0.3 with standard deviation sampled as $0.01 + U(0, 0.01)$. Mixup and stronger corruption recipes are excluded from the default protocol and treated only as appendix diagnostics.

4 Event-Time Modeling Framework

4.1 Scalar and Temporal-Readout Controls

Direct scalar controls. We begin with baselines that treat the task as direct reaction-time regression: a stimulus-locked 2 s EEG crop $X \in \mathbb{R}^{C \times T}$ with $T = 200$ samples (100 Hz) is mapped to a scalar RT. All direct-regression baselines use the fixed 2 s input window, the per-window standardization described above, no mixup, batch size 128, patience 20, and the same release-separated train/development/holdout protocol. The default augmentation is the mild channel-dropout, temporal-cutout, and Gaussian-noise recipe described in the dataset protocol; stronger corruptions are retained only as appendix diagnostics. MSP-CNN denotes a multiscale segment-pooling CNN: it maps the full EEG window to scalar RT through temporal segment statistics and an MLP head, without an explicit representation of response timing. These scalar controls provide reference points for isolating the effect of making temporal structure explicit in Section 4.2; architecture details are collected in Appendix C.

External architectures are evaluated as supervised architectures trained from scratch under the same fixed preprocessing protocol. Pretrained weights are not used. The

Table 1: Model families used in the main comparisons. The paper's main contrast is formulation-level: scalar RT regression and scalar-supervised temporal readout versus event-time posterior supervision.

Family	What it predicts	Main architectural idea	Role in the paper
MSP-CNN	Single scalar RT	Compact temporal ConvNet with multiscale segment-stat pooling and an MLP readout	Direct scalar-regression control; tests the standard window-to-scalar formulation
ETR-CNN	Time-bin scores read out by expectation, but trained with scalar RT supervision	Dilated temporal ConvNet with temporal soft-argmax readout	Temporal-readout control; tests whether a temporal expectation head is enough without event-time distributional supervision
ETS-U-Net	Per-time logits defining an event-time posterior over response latency	Lightweight 1-D U-Net encoder-decoder with skip connections and time-resolved output	Main event-time model family; used for CE, CE+time, time-only, Wasserstein-only, EventNLL, Gaussian-mixture, and hazard variants
Wrapped external EEG backbones	Single scalar RT under the same fixed preprocessing	Standard supervised EEG architectures or foundation-style architectures instantiated from scratch, including EEGNet, Deep4Net, TIDNet, ATCNet, EEGConformer, LaBraM, and EEGPT	Controlled architecture baselines; not a test of pretrained EEG foundation models

main tables report the wrapped/standardized variants because they receive the same per-window standardization as our models; unwrapped external runs are treated as sanity checks in the Appendix.

Temporal-readout controls. Temporal-readout regression is an intermediate control between scalar pooling and full event-time supervision. These models predict scores over time bins and read RT by a temporal expectation, but supervision remains scalar RT or an RT-derived readout loss rather than a full event-time posterior objective. The temporal-readout family includes segment-stat pooling, temporal soft-argmax readout (Nibali et al., 2018), and an ACT-style recurrent readout (Graves, 2016; Perez et al., 2018); representative architecture families are summarized in Appendix Table 9. This control tests whether an expectation-style temporal head alone explains the R11 behavior evaluated in Section 5.1.

Model-family summary. Table 1 summarizes the model names used in the main comparisons. The table is intended as a reader-facing map: it explains what each family predicts and why it is included, while layer-level architectural details remain in Appendix C.

4.2 Event-Time Posterior Formulation

To make event timing explicit and to enable crop augmentation with consistent labels, RT prediction was reformulated as *temporal segmentation*: instead of regressing a scalar directly, we predict a distribution over time bins and read out RT as its expectation (Figure 2). Compared to distributional readouts used only as a final head, the key

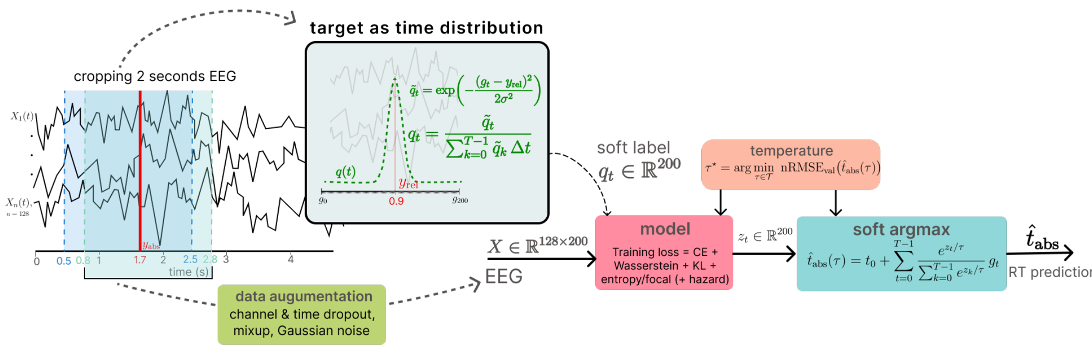

Figure 2: Temporal segmentation pipeline for RT prediction. For each trial we cache a longer stimulus-locked segment (4 s in our implementation) and sample a 2 s subwindow with start-time jitter for training; the RT label is shifted into the sampled window and a Gaussian soft target q_t is formed. Per-time logits z_t are produced by a 1-D segmentation model and are optimized with distribution-matching losses (e.g., soft-label CE, CDF-based Wasserstein-1, KL, and mild entropy penalty) or latent EventNLL objectives. The main readout is a temperature-scaled softmax posterior; hazard/survival readouts and alternate observation kernels are evaluated as compact extensions. RT is obtained by temperature-scaled soft-argmax, with $\tau^\star$ selected to minimize validation nRMSE.

difference here is that supervision is defined in time coordinates and shifted consistently under crop jitter, so augmentation preserves the event-time target by construction. This formulation is designed to align supervision with the event-time nature of the target; the empirical effect is evaluated in Section 5.1.

Notation and crop sampling. Let a 2 s crop be $X \in \mathbb{R}^{C\times T}$ with sfreq $= 100$ Hz, $T = 200$ and $\Delta t = 1/\text{sfreq}$. For training-time augmentation, we retain a longer stimulus-locked segment of length 4 s (0–4 s post-stimulus) and sample a 2 s subwindow from it, introducing temporal jitter while preserving trial identity. The crop start is clipped so that the labeled response remains inside the sampled 2 s window (label-preserving jitter). During validation and official inference, we always use the fixed stimulus-locked 2 s window with offset $t_0 = 0.50$ s (0.50–2.50 s), so that $t_{\text{abs}} = t_0 + t_{\text{rel}}$.

4.3 Event-Time Objectives: CE, CE+Time, Controls, and EventNLL

Soft target distribution. For a sampled crop, the ground-truth event time within the crop is represented as a relative time $y_{\text{rel}} \in [0, \text{crop\_sec})$ with crop_sec $= 2$. A smooth segmentation label is constructed using a Gaussian kernel with bandwidth σ:

$$\tilde{q}_t = \exp\left(-\frac{(g_t - y_{\text{rel}})^2}{2\sigma^2}\right), \qquad q_t = \frac{\tilde{q}_t}{\sum_{k=0}^{T-1} \tilde{q}_k}, \qquad \sum_{t=0}^{T-1} q_t = 1. \tag{1}$$

This smoothing stabilizes optimization while preserving temporal localization; in practice, σ can be annealed (start larger and decrease over training).

Model output and losses. A 1-D segmentation backbone produces logits z_t for $t = 0,\dots,T-1$, and a predicted event-time distribution is obtained via temperature-scaled softmax,

$$p_t(\tau) = \frac{e^{z_t/\tau}}{\sum_{k=0}^{T-1} e^{z_k/\tau}}. \tag{2}$$

The soft-label segmentation family is written as a weighted objective:

$$\begin{aligned}\mathscr{L} &= \lambda_{\text{time}}\,\text{RMSE}\big(\hat{t}_{\text{rel}}(\tau), y_{\text{rel}}\big) + \lambda_{\text{CE}}\,\text{CE}\big(q, p(\tau)\big) \\ &\quad + \lambda_{W1}\,W_1\big(\text{CDF}(q), \text{CDF}(p(\tau))\big) + \lambda_{\text{KL}}\,D_{\text{KL}}\big(q \| p(\tau)\big) - \lambda_H\,H\big(p(\tau)\big),\end{aligned} \tag{3}$$

where

$$\text{CE}(q,p) = -\sum_{t=0}^{T-1} q_t \log p_t,\ D_{\text{KL}}(q\|p) = \sum_{t=0}^{T-1} q_t \log\frac{q_t}{p_t},\ H(p) = -\sum_{t=0}^{T-1} p_t \log p_t.$$

This expression defines an ablation family rather than a single mandatory loss cocktail. CE-only supervision sets the scalar time term to zero and asks the model to match a Gaussian soft event-time label. The CE+time objective combines CE with a soft-argmax time loss. The time-only ablation removes distributional matching while keeping the same readout, and the Wasserstein-only ablation tests whether a CDF-based geometry loss is sufficient on its own. The entropy, KL, focal, and related terms are best viewed as posterior-geometry controls rather than as the main contribution. Gaussian-mixture EventNLL and hazard/survival parameterization are evaluated as compact extensions rather than as separate main contributions.

Latent Event-Time Likelihood. As a probabilistic alternative to soft Gaussian-label supervision, we also evaluate a latent event-time likelihood. The temporal softmax defines a posterior over latent event bins, and the observed scalar RT is modeled as a noisy realization of the latent event time:

$$p(y \mid X) = \sum_{t=0}^{T-1} p_t(X)\,\mathscr{N}\big(y; t_0 + g_t, \sigma_y^2\big), \qquad \mathscr{L}_{\text{EventNLL}} = -\log p(y \mid X). \tag{4}$$

This objective keeps the same event-time interpretation but replaces the artificial target mask q_t with a marginal likelihood over latent response-relevant event times. We therefore present EventNLL primarily as a more principled latent-variable objective, not as the scalar-NRMSE winner among the evaluated losses.

Observation-kernel and hazard extensions. The EventNLL formulation separates the latent event-time posterior $p_t(X)$ from the observation model $K(y \mid t)$. The default model uses a single Gaussian observation kernel, but we also evaluate a two-scale Gaussian mixture,

$$K(y \mid t) = (1-\alpha)\,\mathscr{N}(y; t_0 + g_t, \sigma^2_{\text{narrow}}) + \alpha\,\mathscr{N}(y; t_0 + g_t, \sigma^2_{\text{wide}}), \tag{5}$$

which keeps most probability mass in a narrow timing error while allowing a wider tail for occasional noisy RT observations. In this variant, $\sigma_{\text{narrow}} = 0.10$ s, $\sigma_{\text{wide}} = 0.35$ s, and $\alpha = 0.10$.

As a readout extension, we also evaluate a hazard/survival parameterization. Instead of using logits only as unconstrained softmax scores, each temporal logit defines a conditional event probability

$$h_t = \sigma(z_t/\tau), \qquad p_t^{\text{haz}} \propto h_t \prod_{k<t}(1-h_k), \tag{6}$$

with the resulting event-bin probabilities normalized over the represented window. This keeps the same event-time likelihood in Eq. 4, but changes the posterior parameterization from a categorical softmax to a discrete-time survival process. We treat this as a compact extension that tests whether the event-time formulation is tied to one softmax implementation.

4.4 Readout, Calibration, and Implementation Summary

The posterior view makes geometric properties of the predicted distribution directly inspectable. CDF-based distances penalize temporal displacement of probability mass, entropy and variance summarize posterior sharpness, and quantiles expose asymmetric uncertainty around the predicted response time. These summaries are useful for diagnosis and for downstream stacking.

When used, explicit posterior regularization can be written generically as

$$\mathscr{L} = \mathscr{L}_{\text{event}} + \lambda R(p(t \mid X)), \tag{7}$$

where R may target entropy, variance, tail mass, or other posterior-shape statistics. In this manuscript we treat such terms as analysis and control of posterior geometry unless they are explicitly evaluated as ablations.

Inference and temperature tuning. The relative event time is estimated by expectation (soft-argmax),

$$\hat{t}_{\text{rel}}(\tau) = \sum_{t=0}^{T-1} p_t(\tau)\, g_t, \qquad \hat{t}_{\text{abs}}(\tau) = t_0 + \hat{t}_{\text{rel}}(\tau), \tag{8}$$

where $g_t = t\,\Delta t$ denotes the discrete time grid. Under squared error, the Bayes-optimal point estimate is the conditional mean; accordingly, reading out RT by $\mathbb{E}_{p(t)}[g]$ aligns naturally with the nRMSE objective (RMSE up to a constant normalization).

In practice, different training choices (loss weighting, augmentation strength, and checkpoint selection) produced time-distributions with markedly different sharpness and entropy: some models generated very peaked $p(t)$ while others were systematically more diffuse. This creates a confidence-scale mismatch, where naive averaging (or stacking without temperature-scaled inputs) can underweight accurate but underconfident models and overweight overconfident ones.

We therefore use post-hoc temperature scaling (Guo et al., 2017) to tune the softness of $p(t)$, adjust the readout, and make confidence profiles more comparable across losses and models. Temperature is selected by validation tuning for each base model m:

$$\tau_m^\star = \arg\min_{\tau \in \mathscr{T}_{\mathrm{cal}}} \mathrm{nRMSE}_{\mathrm{val}}\big(\hat{t}_{\mathrm{abs}}^{(m)}(\tau)\big). \tag{9}$$

Under the release-separated protocol, $\mathscr{T}_{\mathrm{cal}}$ is selected on the R9–R10 development split and applied unchanged to R11.

Because confidence scales differ across objectives and architectures, temperature is treated as a posterior-geometry control selected on development data rather than as a separate model-selection signal.

Implementation summary. The main event-time models use the ETS-U-Net family to map a 2 s, 128-channel EEG crop to per-time logits at the original 200-sample resolution. The architecture uses a 1-D encoder-decoder with skip connections, depthwise-separable residual blocks, anti-aliased temporal downsampling, and a dilated bottleneck. Architecture schematics, model-zoo variants, and capacity/attention/factorized checks are reported in Appendix C. Distribution-aware stacking is used only as secondary challenge context and is described in Appendix A. It is not used for the main R11 single-model claims.

5 Experiments and Results

5.1 Main R11 Scalar Accuracy

Table 2 reports direct-regression and external-backbone baselines under the release-separated protocol; the model-family names are summarized in Table 1. The compact MSP-CNN and ETR-CNN models are the strongest direct-regression baselines under the fixed protocol. Wrapped external architectures are substantially more stable under the fixed preprocessing than their unwrapped sanity-check counterparts, but remain behind the compact task-specific baselines on R11.

Temporal-readout control. The temporal-readout ETR-CNN has a slightly lower R11 nRMSE point estimate than the scalar MSP-CNN (0.945938 vs. 0.946269), while increasing ETR-CNN capacity gives a higher held-out score of 0.951909. This suggests that simply parameterizing a temporal expectation is not sufficient; the larger point-estimate gap appears when the time distribution itself is supervised.

Event-time segmentation. Table 3 summarizes the event-time segmentation loss ablations. The calibrated CE+time event-time segmentation model reaches 0.935573 nRMSE on R11, a lower point estimate than the strongest direct-regression baseline.

Table 2: Direct-regression and wrapped external backbone baselines under the release-separated protocol. R11 scores are reported with subject-bootstrap confidence intervals. Lower nRMSE is better.

Model	Role	Valid nRMSE ↓	R11 nRMSE ↓
MSP-CNN	compact scalar regression baseline	0.933637	0.946269 [0.931609, 0.962943]
ETR-CNN	temporal-readout direct regression	0.932577	0.945938 [0.930153, 0.963795]
ETR-CNN large	capacity ablation	0.939308	0.951909 [0.936078, 0.969803]
TIDNet wrapped	strongest external supervised baseline in this comparison	0.957974	0.959004 [0.948507, 0.970822]
EEGConformer wrapped	modern supervised transformer-style baseline	0.957968	0.962802 [0.951634, 0.975382]
EEGNet wrapped	canonical compact EEG baseline	0.957877	0.965041 [0.954684, 0.977111]
LaBraM wrapped	foundation-style architecture from scratch	0.958254	0.965382 [0.952576, 0.979330]
Deep4Net wrapped	classical convolutional baseline	0.955678	0.971619 [0.956680, 0.989067]
ATCNet wrapped	supervised conv/attention/TCN baseline	0.970046	0.977366 [0.966497, 0.990685]
EEGPT wrapped	foundation-style architecture from scratch	0.967140	0.978796 [0.969367, 0.990273]

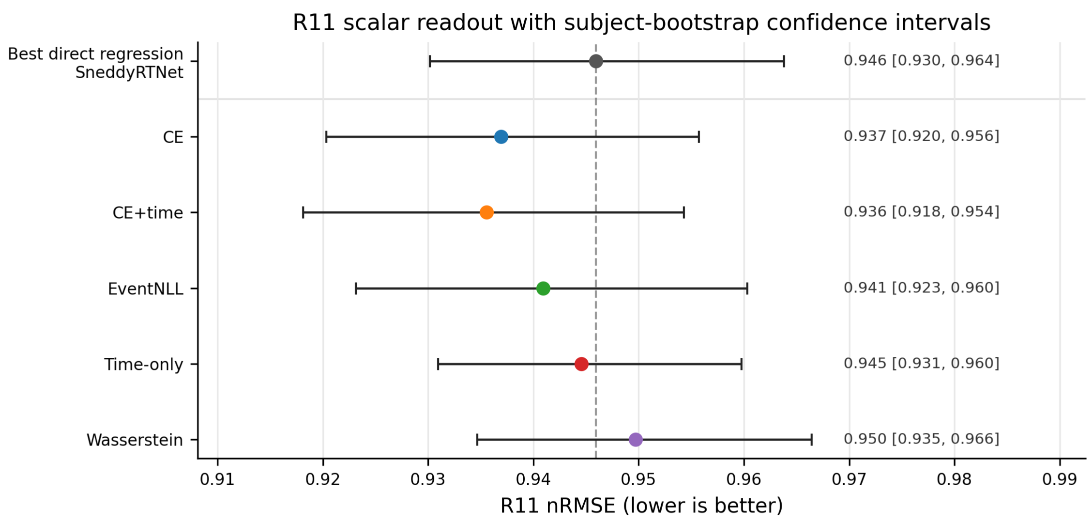

Figure 3: R11 scalar readout performance with subject-bootstrap confidence intervals. The dashed vertical line marks the strongest direct-regression baseline. Several event-time losses have lower scalar nRMSE point estimates while remaining close enough to motivate posterior-level analysis rather than ranking losses only by point-prediction error.

CE-only soft-label supervision is also strong. The Gaussian-mixture and hazard rows are compact EventNLL-family extensions: the former changes the observation kernel, while the latter changes the event-time posterior parameterization. This is the central empirical step in the narrative: once RT is represented as an event-time posterior, distributional supervision over time becomes more useful than treating RT as a scalar window label.

5.2 Event-Time Supervision and Loss Ablations

The loss ablations separate the value of the temporal readout from the value of distributional supervision. The time-only model directly optimizes soft-argmax error but

Table 3: Event-time segmentation loss ablations under the release-separated protocol. "Cal." denotes post-hoc temperature selection on R9–R10 and one-shot application to R11.

Run	Objective	τ	Valid nRMSE ↓	R11 nRMSE ↓
CE only, base	soft-label event-time CE	0.65	0.922529	0.938738 [0.921696, 0.958015]
CE only, cal.	soft-label event-time CE	0.70	0.922277	0.936942 [0.920329, 0.955704]
Hybrid, base	soft-label CE + soft-argmax time loss	0.65	0.925601	0.940006 [0.921243, 0.960020]
Hybrid, cal.	soft-label CE + soft-argmax time loss	0.80	0.924544	0.935573 [0.918136, 0.954309]
Event NLL, cal.	continuous latent event-time likelihood	0.70	0.923619	0.940947 [0.923160, 0.960322]
Gaussian mixture NLL, cal.	two-scale EventNLL observation kernel	0.75	0.921676	0.939267 [0.921864, 0.958291]
Hazard EventNLL	hazard-parameterized event-time posterior	0.65	0.924318	0.937332 [0.920594, 0.955704]
Time only, cal.	soft-argmax scalar time loss only	0.85	0.935298	0.944578 [0.930935, 0.959775]
Wasserstein only, cal.	event-time CDF distance only	2.95	0.936941	0.949716 [0.934666, 0.966443]

removes distribution matching; it is weaker than CE-only and hybrid segmentation. The Wasserstein-only run is a useful negative control: not every distributional distance recovers the gain. CE, CE+time, and EventNLL remain in the same performance band, supporting the view that the main gain comes from the event-time formulation rather than from a single hand-tuned loss. The hazard EventNLL result further shows that the formulation is not tied to a categorical softmax readout: a survival-style posterior reaches the same scalar-performance regime while preserving the event-time likelihood view.

EventNLL as a principled likelihood. The event-time NLL run nearly matches CE-only and hybrid segmentation without requiring a hand-weighted CE+time objective. We therefore present it as a principled latent-variable likelihood rather than as a claim of best scalar nRMSE. The Gaussian-mixture kernel has a lower EventNLL-family scalar-readout point estimate than the fixed Gaussian kernel, while the hazard EventNLL extension reaches CE-level R11 performance without reverting to scalar regression. In the calibrated posterior analysis, EventNLL produces a much narrower posterior than CE or CE+time and places more mass near the target, but it also under-covers nominal central intervals. This makes it scientifically useful: it reveals that similar scalar errors can correspond to different posterior geometries.

5.3 Seed Robustness

To test whether the main comparison depends on a favorable random initialization, we repeated selected core recipes over five seeds (2025–2029). This analysis is a reliability check rather than a new model-selection stage: it focuses on the strongest temporal-readout regression baseline, the strongest wrapped external baseline, the two main distributional event-time objectives, and the time-only segmentation control.

Table 4 shows that the central event-time supervision gap is stable. CE and EventNLL segmentation remain about 0.010–0.011 R11 nRMSE lower than the repeated ETR-CNN scalar baseline on average. In contrast, the time-only segmentation control is much

Table 4: Five-seed robustness summary for selected core models. Values are mean ± sample standard deviation over seeds 2025–2029; R11 range is min–max across seeds. Posterior scores are reported only for event-time segmentation runs. Lower is better for all metrics.

Model	Role	Valid nRMSE ↓	R11 nRMSE ↓	R11 range	Posterior CRPS ↓	Fixed NLL ↓
ETR-CNN	temporal-readout scalar baseline	0.9359 ± 0.0031	0.9495 ± 0.0048	0.9451–0.9570	–	–
TIDNet wrapped	strongest wrapped external baseline	0.9546 ± 0.0014	0.9614 ± 0.0046	0.9547–0.9656	–	–
ETS-U-Net CE	soft-label event-time CE	0.9209 ± 0.0024	0.9384 ± 0.0021	0.9352–0.9407	0.1927 ± 0.0007	0.3062 ± 0.0086
ETS-U-Net EventNLL	latent event-time likelihood	0.9242 ± 0.0024	0.9391 ± 0.0045	0.9318–0.9439	0.2011 ± 0.0019	0.1523 ± 0.0133
ETS-U-Net time-only	soft-argmax time loss control	0.9371 ± 0.0020	0.9473 ± 0.0021	0.9447–0.9496	0.2040 ± 0.0020	0.3878 ± 0.0268

Table 5: Quantitative posterior geometry on R11 for matched event-time segmentation losses. Scalar nRMSE and MAE summarize point-readout accuracy; CRPS and fixed-kernel EventNLL are distributional scores computed from the full posterior; width, near-target mass, mode-mean gap, and empirical coverage quantify posterior concentration and calibration. CRPS is reported in milliseconds. Fixed-kernel EventNLL uses the same Gaussian observation kernel for all models ($\sigma = 0.15$ s), so lower values indicate higher likelihood of the observed RT under the predicted event-time mixture; because this is a continuous negative log-density, values can be negative. Coverage MAE is the mean absolute deviation between empirical and nominal central-interval coverage across evaluated nominal levels.

Model	nRMSE	MAE ms	CRPS ms	Fixed NLL	Width80 ms	Mass ±150 ms	Mode-mean gap ms	Coverage80	Coverage MAE
CE	0.937	216	155	0.115	820	0.328	57	0.895	0.113
CE+time	0.936	219	157	0.144	860	0.311	57	0.899	0.116
EventNLL	0.941	216	162	-0.027	450	0.487	72	0.480	0.290
Time-only	0.945	228	167	0.203	1070	0.311	129	0.913	0.113
Wasserstein	0.950	222	165	0.219	740	0.296	50	0.812	0.053

closer to ETR-CNN, supporting the interpretation that the gain comes from distributional event-time supervision rather than only from a U-Net backbone or soft-argmax readout. The optional repeated TIDNet baseline also has higher average R11 nRMSE than both ETR-CNN and the event-time segmentation variants.

5.4 Posterior Geometry

Temperature calibration is applied before comparing scalar readouts and posterior summaries. Figure 4 summarizes posterior geometry metrics, and Figure 5 shows trial-sorted posterior maps across the R11 representable subset. Together, these views show that losses with similar scalar nRMSE can learn very different temporal posteriors. EventNLL is sharply target-aligned but under-covers, time-only posteriors are broader and show larger mode-mean gaps, and CE/CE+time provide better empirical coverage. This supports treating posterior sharpness as a first-class analysis object: the model's probability geometry affects scalar readout, uncertainty interpretation, and the reliability of downstream ensembling.

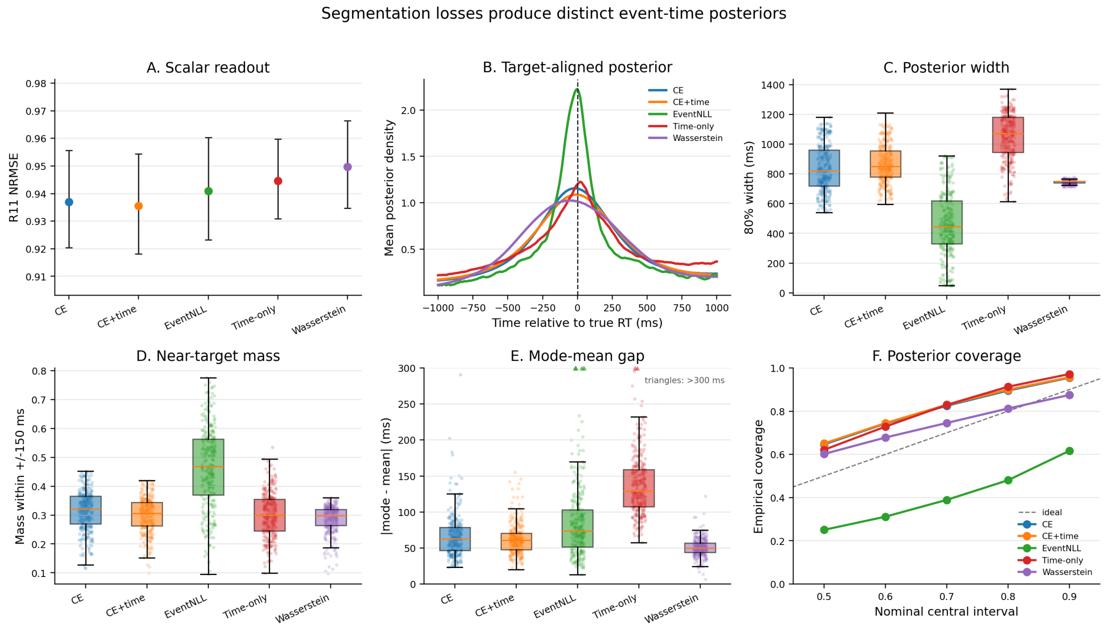

Figure 4: Posterior geometry differs across event-time losses even when scalar nRMSE is similar. Panel A reports calibrated R11 scalar readout with subject-bootstrap intervals. Panel B aligns predicted posteriors to the true RT. Panels C–E summarize posterior width, near-target mass, and mode-mean disagreement. Panel F compares empirical coverage of central posterior intervals. EventNLL is a principled latent-time likelihood and produces sharper target-aligned posteriors, but it is not the best scalar-NRMSE model and under-covers relative to CE-based objectives.

5.5 Shifted-Crop Diagnostic

The posterior analyses above show that event-time losses produce different probability geometries, but they do not by themselves prove that a model has learned a within-crop event localizer. A fixed stimulus-locked window can still permit shortcut behavior: a model may predict trial difficulty, subject-level slowness, or stimulus-locked response tendency without moving its prediction when the crop is shifted. We therefore introduced a shifted-crop diagnostic on R11. For each trial, we used a 5 s EEG window and evaluated the same trained model on 2 s crops starting at $s \in \{0.2, 0.3, \ldots, 0.8\}$ s. The reference crop is $s = 0.5$ s, matching the training and main inference protocol. To avoid target-censoring effects, this analysis uses the common-inside subset where RT remains inside all shifted crops ($0.8 \leq \mathrm{RT} \leq 2.2$ s; 14,183 trials).

If the model truly localizes the response-relevant event within the crop, shifting the crop start by Δs should move the predicted crop-relative event time by approximately $-\Delta s$, giving a raw shift slope near -1. A crop-invariant scalar regressor should instead have a slope near 0. Table 6 shows that the event-time segmentation models have lower reference-crop nRMSE point estimates and move in a more localizer-like direction than the best scalar regressor. EventNLL reaches a raw shift slope of -0.292 [95% CI: -0.311, -0.272], compared with -0.253 [-0.270, -0.235] for ETR-CNN and -0.172

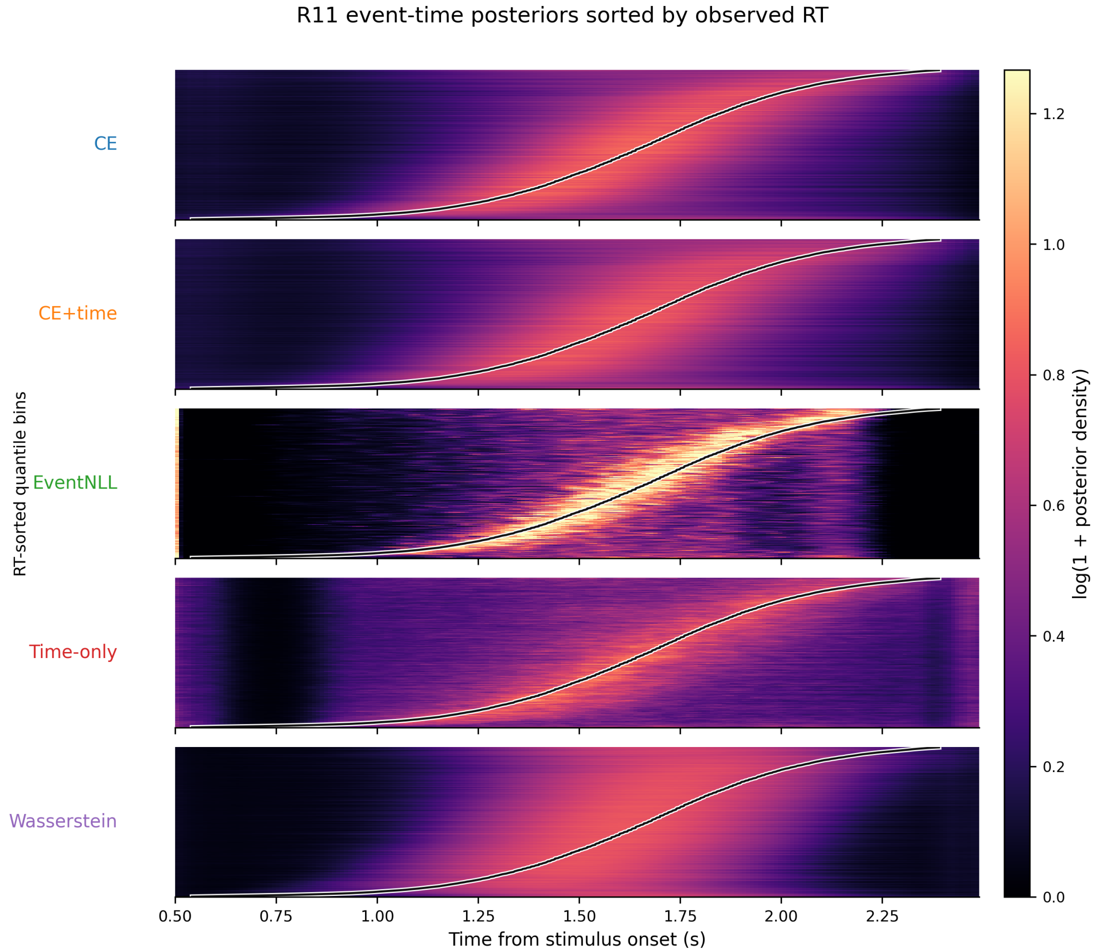

Figure 5: Trial-sorted event-time posterior maps on R11 for matched segmentation losses. Rows are quantile bins of representable R11 trials sorted by observed RT; the x-axis is time from stimulus onset; color shows log-transformed posterior density averaged within each bin. The black curve marks the mean observed RT in each bin.

on average across regression baselines. The fraction of localizer-like trials also increases from 10.1% for ETR-CNN to 14.8% for CE and 15.9% for EventNLL.

This result should be interpreted as evidence for a representational shift, not as solved shift equivariance. The segmentation models remain far from the ideal slope of -1, indicating that fixed-window training still mixes event localization with scalar response-tendency shortcuts. That limitation is useful: it identifies shift-augmented, crop-relative event-time training as the natural next step if the goal is a stronger trial-wise neural event localizer.

Table 6: Shifted-crop diagnostic on the R11 common-inside subset. The ideal within-crop event localizer has raw shift slope -1; a crop-invariant scalar shortcut has slope 0. Reference nRMSE is computed at the standard $s = 0.5$ s crop. Localizer-like trials have per-trial raw slope in $[-1.25, -0.75]$; invariant-like trials have per-trial raw slope in $[-0.25, 0.25]$.

Model	Ref nRMSE ↓	Ref MAE, s ↓	Worst Δ nRMSE	Raw shift slope	Localizer-like trials	Invariant-like trials
ETR-CNN	0.874	0.206 [0.199, 0.215]	0.174	−0.253 [−0.270, −0.235]	10.1% [9.0, 11.1]	34.5%
ETS-U-Net CE	0.856	0.201 [0.193, 0.209]	0.193	−0.287 [−0.306, −0.267]	14.8% [13.3, 16.3]	34.4%
ETS-U-Net EventNLL	0.859	0.201 [0.193, 0.210]	0.198	−0.292 [−0.311, −0.272]	15.9% [14.5, 17.3]	33.3%

6 Discussion

6.1 Interpretation and Relation to EEG Backbones

The results support treating trial-wise RT prediction as event-time localization rather than as ordinary scalar regression. The label is the latency of a response within a post-stimulus interval, and models benefit when that timing structure is explicit in both the output space and the loss. The shifted-crop diagnostic adds a stricter check on this interpretation: event-time segmentation models move in a more event-like direction than scalar regression under temporal crop shifts, but their slopes remain far from perfect localizer behavior. This framing therefore connects naturally to response-locked EEG analyses while also making clear that fixed-window RT prediction is not the same as fully shift-equivariant event detection.

Distributional supervision. Distributional supervision gives the model a local temporal target instead of a single global scalar penalty. CE-only and hybrid losses outperform the time-only loss, suggesting that the gain is not just a differentiable soft-argmax readout. The soft event-time target encourages probability mass near plausible response latencies and makes label-consistent temporal crop jitter easier to justify. EventNLL reaches a similar scalar-error regime through a different route: it marginalizes a latent event time instead of asking the model to imitate a hand-constructed Gaussian mask.

Backbones and pretraining. The wrapped external baselines show that standard EEG architectures can train under the fixed normalization protocol, but they do not close the gap to compact task-specific models or event-time segmentation. Foundation-style architectures are evaluated here as architectures trained from scratch, not as pretrained models. The comparison is therefore not a test of pretraining; it shows that, under this benchmark protocol, formulation and objective design are strong levers even with lightweight supervised models.

6.2 What the Posterior Adds Beyond Scalar RT Prediction

The event-time posterior contains more information than its expectation. Sharpness, entropy, variance, quantiles, and the gap between posterior mode and posterior mean describe confidence and ambiguity in ways that scalar predictions discard. The posterior analysis shows that CE, CE+time, EventNLL, time-only, and Wasserstein losses can have similar scalar readouts but different target-alignment, width, and coverage behavior. Temperature calibration is therefore best understood as posterior-geometry control and a fairness step for comparing readouts, not as the main methodological claim.

6.3 Limitations and Future Work

The main limitation is that all claims are tied to HBN-EEG CCD and the release-separated protocol used here. R11 is a stronger local holdout than reused validation performance, but it is still drawn from the same broader dataset family and preprocessing pipeline. The shifted-crop diagnostic shows partial localization, not full shift equivariance, so stronger claims about shift-equivariant trial-wise event localization would require training protocols that explicitly randomize crop position and supervise crop-relative timing. The five-seed robustness checks support the main scalar-versus-event-time comparison, but they cover selected core recipes rather than every kernel, hazard, calibration, and architecture variant. The foundation-style comparisons also do not use pretrained weights or montage-specialized setup changes; they are controlled architecture baselines rather than a full evaluation of EEG foundation models.

7 Conclusion

We reformulated EEG-based reaction-time prediction as event-time distribution modeling and evaluated it under a release-separated HBN-EEG protocol. The results indicate that supervising an event-time posterior gives lower R11 nRMSE point estimates than scalar regression and a temporal-readout head alone. Loss ablations and five-seed robustness checks show that distributional event-time supervision matters, EventNLL provides a principled latent-variable likelihood alternative, and posterior-geometry plus shifted-crop analyses reveal output behavior that scalar nRMSE alone cannot capture.

A Original Hidden CodaBench and Stacking

We report the original hidden-release CodaBench evaluation as historical challenge context. These scores provide continuity with the original benchmark setting, while the main empirical conclusions in this manuscript are based on the labeled R11 release with subject-bootstrap confidence intervals.

The R11 results in the main text are intentionally reported without post-hoc stacking, so that the main comparisons isolate the event-time objective, scalar readout, posterior geometry, and shifted-crop behavior. The available stacking evidence comes from the original hidden-release CodaBench evaluation in Table 7. In that earlier challenge pipeline, gradient boosting stacking achieved 0.91394 hidden nRMSE, compared with 0.92334 for the best single model, 0.91840 for manual blending, and 0.91539 for Ridge

Table 7: Original hidden-release CodaBench RT results from the earlier challenge pipeline. These results are reported for historical comparison with the challenge setting.

Method	Validation nRMSE ↓	Hidden nRMSE ↓
Best temperature-tuned single model	0.898757	0.92334
Manual blending	0.899979	0.91840
Ridge stacking	0.890430	0.91539
Gradient boosting stacking	**0.883365**	**0.91394**

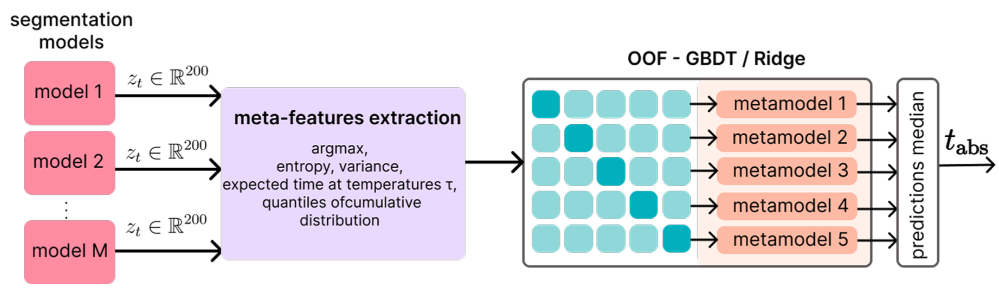

Figure 6: OOF stacking for RT prediction from time-distribution outputs. Per-model logits/distributions are converted into meta-features (e.g., peak time, entropy/variance, temperature-conditioned expected times, and CDF quantiles) and combined by a second-stage stacker (Ridge/GBDT). Predictions from fold-trained stackers are aggregated at inference (median) to produce the final RT estimate.

stacking. This supports the role of posterior-derived meta-features as an ensemble calibration layer, while the release-separated R11 claims remain based on the single-model comparisons and diagnostics in Tables 2–6.

A.1 Distribution-Aware Stacking

Segmentation-style outputs enable richer ensembling than scalar averaging, but combination becomes sensitive to model confidence: naive averaging can overweight overconfident models and underweight accurate but underconfident ones. We therefore define an auxiliary second-stage stacker on distributional meta-features extracted from the time-distribution outputs.

In a release-separated application, base segmentation models are fit on R1–R8 before the stacker is trained. Meta-model inputs are then computed on the R9–R10 development partition using subject-disjoint folds, so that the stacker never receives in-fold predictions for the same subject. Logits and temperature-scaled event-time probabilities are converted to meta-features such as peak time, entropy, variance, expected time, and CDF quantiles. This stacker is treated as a post-hoc ensemble layer rather than as part of the single-model event-time comparison.

Meta-features from logits and temperature-scaled probabilities. Meta-features are extracted per model from (i) raw logits and (ii) temperature-scaled probabilities

Table 8: Unwrapped external-backbone sanity checks. “Wrapped” means that the model receives the same per-window channel standardization used by our compact baselines. Lower nRMSE is better. These checks diagnose scale sensitivity; the wrapped variants are the fair external comparisons used in the main results.

Architecture	Unwrapped valid	Unwrapped R11	Wrapped valid	Wrapped R11	Takeaway
EEGNet	1.000000	1.002046 [0.999718, 1.007253]	0.957877	0.965041 [0.954684, 0.977111]	Standardization makes the compact EEG baseline train
EEGConformer	1.000000	1.000413 [0.999906, 1.003179]	0.957968	0.962802 [0.951634, 0.975382]	Transformer-style baseline is scale-sensitive without wrapping
LaBraM	1.000000	1.000711 [0.998638, 1.004512]	0.958254	0.965382 [0.952576, 0.979330]	From-scratch foundation-style architecture needs the fixed normalization

computed for multiple temperatures $\tau \in \mathscr{T}_{\text{meta}}$. From logits, a hard-peak estimate and confidence proxies are computed:

$$t^\star = \arg\max_t z_t, \qquad \hat{t}_{\text{hard}} = t_0 + g_{t^\star}, \qquad z_{\max} = \max_t z_t, \qquad m_{\text{top2}} = z_{\max} - \max_{t \neq t^\star} z_t. \tag{10}$$

For each $\tau \in \mathscr{T}_{\text{meta}}$, probabilities are formed as $p(\tau) = \text{softmax}(\mathbf{z}/\tau)$ (Eq. 2, Sec. 4.2), where $\mathbf{z} = (z_0, \dots, z_{T-1})$ denotes the per-time logit vector for a given model and window. Distributional summaries are computed, including the soft-argmax time (Eq. 8, Sec. 4.2), entropy $H(p(\tau))$, peak probability $p_{\max}(\tau)$, time variance $\text{Var}_{p(\tau)}[g]$, and CDF time-quantiles (10/50/90%). In our implementation, $\mathscr{T}_{\text{meta}} = \{0.7, 1.0, 1.3\}$.

Meta-learner and inference. We use linear Ridge regression as a strong baseline meta-learner, and gradient boosting to capture non-linear reliability patterns across models and temperatures. The meta-learner is trained using subject-disjoint cross-validation: each subject is assigned to exactly one fold (Figure 6). For fold k, the stacker is fit on the other folds and predicts RT for the held-out fold. The leakage-controlled diagnostic is the out-of-fold nRMSE from these held-out predictions. Folds are stratified to match the per-subject mean RT distribution across folds. For held-out inference, the same feature construction is applied to unseen logits; predictions from the fold-trained stackers are aggregated to obtain the final RT estimate. Manual blending serves as a reference, but it cannot adapt weights to sample-specific confidence profiles.

B External-Backbone Sanity Checks

Unwrapped external architectures are useful for diagnosing scale sensitivity. Because our own models use per-window standardization, the primary comparison uses wrapped external models. Table 8 reports matched sanity checks for representative off-the-shelf architectures with and without the fixed per-window standardization wrapper. Without the fixed benchmark normalization, some off-the-shelf instantiations were scale-sensitive and often collapsed toward mean-like predictions (nRMSE near 1), whereas the wrapped variants train meaningfully under the same release-separated protocol used in the main tables.

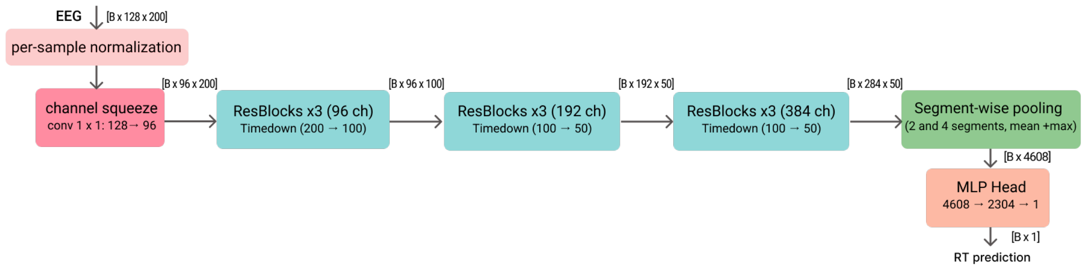

Figure 7: MSP-CNN direct RT regression baseline architecture. A lightweight 1-D temporal ConvNet maps a stimulus-locked 2 s EEG crop (128 channels, 100 Hz) to a scalar reaction-time prediction. Per-sample normalization and a learned 1×1 channel-squeeze layer (128→96) are followed by residual depthwise-separable stages with anti-aliased temporal downsampling, segment-wise pooling (2 and 4 segments; mean+max), and an MLP head.

C Architecture and Implementation Details

C.1 Direct-Regression Architecture

The main MSP-CNN regression backbone (Figure 7) is a lightweight 1-D temporal ConvNet with: (i) per-sample normalization to reduce cross-subject amplitude variation, (ii) a learned channel-squeeze layer (1×1 conv, 128→96) that forms data-driven electrode mixtures, and (iii) an EfficientNet-like temporal hierarchy with widening stages of residual depthwise-separable blocks (96→192→384 channels) and anti-aliased temporal downsampling. Coarse timing cues are retained using segment-wise statistical pooling: the time axis is split into 2 and 4 segments, mean and max are computed per segment, and the resulting features are passed to a small MLP head.

C.2 ETS-U-Net Architecture

The ETS-U-Net family is used to map a stimulus-locked EEG crop $X \in \mathbb{R}^{C \times T}$ to per-time logits $z_{0:T-1}$ (Figure 8). All segmentation variants follow a 1-D U-Net style encoder-decoder with skip connections for precise localization (Ronneberger et al., 2015). The encoder progressively downsamples the temporal axis to build multi-scale context, while the decoder upsamples and fuses encoder features to recover per-time outputs at the original 200-sample resolution (internal down/up-sampling is used only to form multi-scale features). Each scale is refined with lightweight residual blocks (ResNet-style skips) (He et al., 2016) built from depthwise-separable 1-D convolutions for computational efficiency (Howard et al., 2017; Chollet, 2017). To mitigate aliasing under temporal decimation, downsampling applies a fixed depthwise smoothing filter followed by strided averaging ("blur-pool" style) (Zhang, 2019). A dilated bottleneck further enlarges the receptive field without additional downsampling (Yu and Koltun, 2016).

Segmentation model zoo. Table 10 lists the variants used for ensembling/stacking. ETS-U-Net-L/M/S share the same ETS-U-Net template and vary width and the number of

Table 9: Regression-style architecture families used before the event-time segmentation pivot. *DS* = depthwise-separable. Current release-separated results are reported in Section 5.1.

Model	Key design (high-level)	Temporal readout	Manuscript role	Size (MB)
MSP-CNN-S	**Compact DS-residual encoder** Shallow depth and small kernels emphasize local temporal motifs. Strong temporal compression yields coarse features suited to pooling.	Segment-stat pooling + MLP	compact MSP-CNN baseline	4.95
MSP-CNN-L	**Higher-capacity DS-residual encoder** Wider/deeper stack and larger kernels increase temporal context and channel capacity. Same coarse temporal hierarchy, but with stronger representational power.	Segment-stat pooling + MLP	capacity ablation	45.66
ETR-CNN-Dilated	**Dilated DS-residual backbone** Dilations expand receptive field while delaying temporal compression relative to MSP-CNN. Designed to support distributional timing rather than coarse pooling.	Temporal soft-argmax	ETR-CNN baseline	49.24
Recurrent ETR-CNN (ACT)	**Tied-weight recursive DS core** Iterative refinement with step conditioning; adaptive computation via ACT halting. Large effective depth at low parameter cost, paired with distributional timing.	ACT temporal soft-argmax	exploratory readout variant	1.67

residual refinement blocks per level. Attention ETS-U-Net replaces the bottleneck with temporal multi-head self-attention (MHSA) (Vaswani et al., 2017) and uses stochastic depth (DropPath) regularization (Huang et al., 2016) to stabilize training at higher capacity. EEG-Inception Segmentation replaces single-path temporal filtering with Inception-style multi-branch temporal kernels (Szegedy et al., 2015), following the EEG-Inception design for ERP feature extraction (Santamaría-Vázquez et al., 2020). Finally, Factorized ETS-U-Net augments the channel-mixing front-end with factorized cross-channel interactions inspired by Factorization Machines (FM) (Rendle, 2010), optionally also injected within encoder stages to introduce a distinct cross-channel inductive bias.

Architectural diversity is introduced to support ensembling and stacking. ETS-U-Net-L (wide) uses a wider encoder and 5 residual blocks per level, providing the highest capacity. ETS-U-Net-M retains the same depth (5 blocks/level) with a more moderate width for a stronger accuracy–efficiency trade-off, while ETS-U-Net-S uses a lighter encoder (3 blocks/level) and yields faster, lower-variance base learners. Long-range temporal interactions are emphasized in Attention ETS-U-Net (MHSA) via a multi-head self-attention bottleneck. EEG-Inception Segmentation replaces single-path temporal

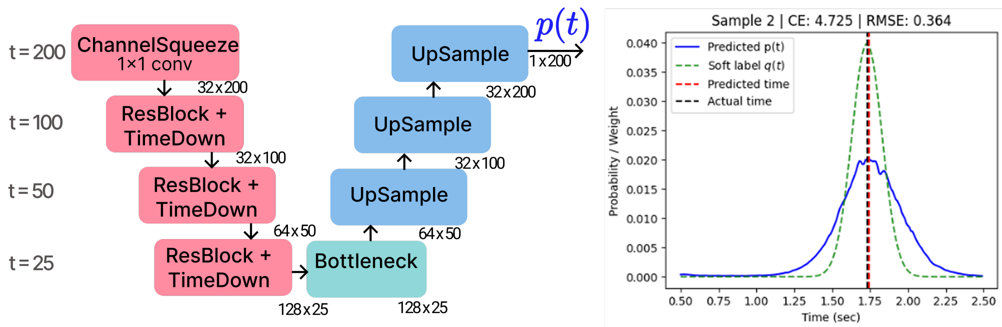

Figure 8: ETS-U-Net for RT segmentation. A lightweight 1-D U-Net maps a 2 s, 128-channel EEG crop to per-time logits and an event-time distribution $p(t)$. The encoder applies a 1×1 channel-squeeze layer followed by residual blocks with temporal down-sampling (`TimeDown`); a bottleneck aggregates context, and the decoder upsamples with skip connections to recover time-resolved outputs. The right panel shows an example prediction $p(t)$ and Gaussian soft target $q(t)$, together with the resulting predicted vs. true response time.

Table 10: Segmentation model variants used for reaction-time prediction. Each model outputs per-time logits and is trained under the same segmentation objective; architectural diversity is exploited for ensembling/stacking.

Model	**Architecture**	**Key features**	**Strengths**
ETS-U-Net-L (wide)	ETS-U-Net	wider encoder; depthwise-separable convs; 5 ResBlocks/level; dilated bottleneck	highest capacity; captures complex local temporal patterns
ETS-U-Net-M	ETS-U-Net	moderate width; depthwise-separable convs; 5 ResBlocks/level; dilated bottleneck	best single-model accuracy / efficiency tradeoff
ETS-U-Net-S	ETS-U-Net	lighter encoder; depthwise-separable convs; 3 ResBlocks/level; same bottleneck family	fast, low-variance; complements deeper models
Attention ETS-U-Net (MHSA)	Attention ETS-U-Net	MHSA bottleneck (4 heads, depth=3); DropPath; positional depthwise conv	captures long-range temporal dependencies
EEG-Inception Segmentation	EEG-Inception 1D	multi-branch temporal scales (0.5/0.25/0.125 s) with feature fusion	strong multi-scale feature extraction
Factorized ETS-U-Net	Factorized ETS-U-Net	FM-style cross-channel interactions at input; optional per-stage FM injection	distinct cross-channel inductive bias

filtering with multi-branch kernels for multi-scale feature extraction. Finally, Factorized

Table 11: Additional EventNLL kernel and hazard robustness checks. Lower nRMSE is better. "Cal." denotes post-hoc temperature selection on R9–R10; the heteroscedastic run is reported at its base readout because it was not part of the calibrated kernel table.

Variant	Change tested	τ	Valid nRMSE ↓	R11 nRMSE ↓	Takeaway
Gaussian EventNLL, cal.	fixed Gaussian observation kernel	0.70	0.923619	0.940947 [0.923160, 0.960322]	Core latent event-time likelihood reference
Gaussian mixture EventNLL, cal.	narrow/wide Gaussian observation mixture	0.75	0.921676	0.939267 [0.921864, 0.958291]	Strongest calibrated EventNLL-family scalar readout
Laplace EventNLL, cal.	Laplace observation kernel	0.75	0.926636	0.940696 [0.923718, 0.959285]	Robust kernel did not yield a lower point estimate than cleaner Gaussian/mixture variants
Student-t EventNLL, cal.	heavy-tailed Student-t observation kernel	0.80	0.923259	0.941127 [0.923382, 0.960687]	Calibration recovers validation gap, but R11 remains tied or weaker
Heteroscedastic EventNLL	trial-wise learned observation scale $\sigma(x)$	0.65	0.923056	0.942343 [0.923630, 0.962703]	Learned scale did not lower R11 nRMSE enough for the core table
Hazard EventNLL	hazard posterior with continuous Gaussian EventNLL	0.65	0.924318	0.937332 [0.920594, 0.955704]	Strong alternative posterior parameterization
Hazard discrete NLL	exact-bin discrete survival likelihood	0.65	0.926288	0.946823 [0.928204, 0.967220]	Negative control: exact-bin survival loss is weaker than continuous EventNLL

ETS-U-Net augments the standard channel-mixing front-end with an FM-style factorized interaction module (optionally also injected within encoder stages), providing a distinct cross-channel inductive bias.

C.3 Additional Architecture Ablations

Architecture ablations separate formulation effects from capacity and backbone-specific effects. In particular, ETS-U-Net capacity variants, attention bottlenecks, factorized channel interactions, recurrent variants, and EEG-Inception-style segmentation provide complementary checks on whether the event-time formulation depends on a single encoder-decoder instantiation.

D Additional EventNLL Kernels and Hazard Checks

The main text keeps the event-time segmentation table focused on the core CE, EventNLL, and control objectives, plus two compact EventNLL-family extensions. Table 11 reports the broader kernel and hazard robustness checks. These runs use the same release-separated protocol and test whether the probabilistic event-time result depends on one specific Gaussian observation kernel or one specific softmax parameterization.

Overall, these checks support two points. First, the latent event-time view is not tied to a single Gaussian-softmax implementation: the hazard EventNLL and Gaussian-mixture kernel remain competitive. Second, more robust or more flexible observation models are not automatically better; Laplace, Student-t, heteroscedastic, and exact-bin hazard variants are useful robustness probes but do not strengthen the core R11 story enough to feature in the main table.

E Additional Protocol Calibration and Augmentation Diagnostics

Protocol calibration evaluated optimizer choice, learning rate, soft-label width, batch size, and augmentation strength before fixing the training recipe used in the release-separated experiments. These diagnostics informed the protocol but are separated from the R11 results used for the main claims.

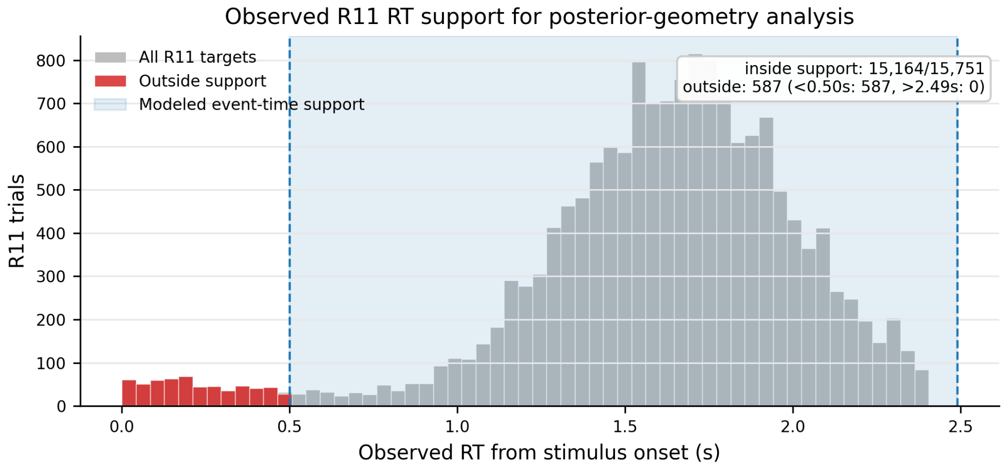

Figure 9: Observed R11 reaction-time distribution relative to the modeled event-time support. Scalar nRMSE is computed on all R11 trials, whereas posterior-geometry summaries are restricted to trials whose observed RT falls inside the event-time grid.

F Additional Posterior-Geometry Examples

Table 5 reports the quantitative posterior metrics behind Figure 4. Figure 10 shows representative trials where models have similar scalar predictions but different posterior widths, large mode-mean gaps, or strong disagreement across losses, and Figure 11 shows the development-split temperature curves used to choose post-hoc readout temperatures.

Figure 10: Representative calibrated event-time posteriors. Dashed black lines denote the true RT, and dotted colored lines denote model-specific posterior means. The examples illustrate why scalar RT alone is insufficient: models can agree on the mean while assigning very different posterior width, or disagree strongly because the posterior is multimodal or skewed.

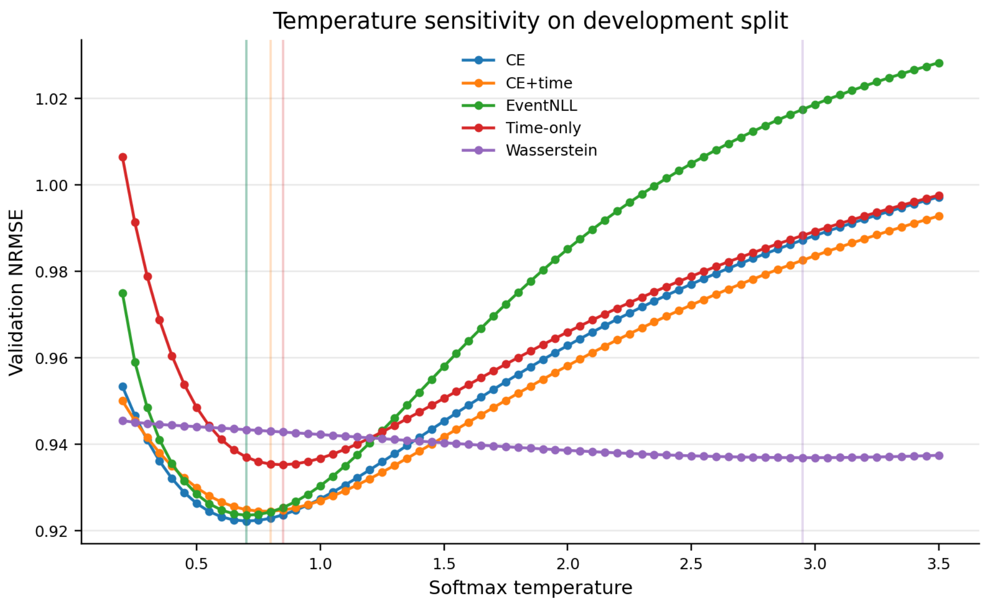

Figure 11: Temperature sensitivity on the R9–R10 development split. Vertical lines mark selected temperatures. The curves show that different losses induce different confidence scales, motivating temperature calibration as posterior-geometry control before comparing scalar readouts or using posterior features downstream.

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